PHYS3071

Introduction to Astrophysics

Course Notes

Author: *automatoc-ti*

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Introduction

About These Notes

These notes accompany the PHYS3071 Astrophysics course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in basic physics, basic linear algebra, and basic calculus.

The material covers different astrophysical topics, including but not limited to the formation and evolution of stars, the type and spectrum of stars. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

If you find any errors or have suggestions for improvement, please feel free to contact me at `yhip@connect.ust.hk`. You can check the latest version of the notes on GitHub: https://github.com/AuTomatoc-Ti/phys3071-Astrophysics-notes/blob/main/main.pdf.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

PHYSICS INTUITION

Purpose: Builds conceptual understanding through plain language explanations.
This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

Theory

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

Question Examples

Purpose: Demonstrates practical application through worked problems.
Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

Foundation

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.
This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

Extra

Purpose: Provides supplementary material that extends *beyond* the course syllabus.
This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

IMPORTANT NOTE

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.
An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

REMARK

Purpose: Highlights a useful observation, connection, or naming convention.
A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

Key symbols.

Symbol	Meaning	Typical units
$M_\odot$	Solar mass	kg
$R_\odot$	Solar radius	m
$L_\odot$	Solar luminosity	W
T_{eff}	Effective temperature	K
μ	Mean molecular weight	—
ρ	Mass density	$\mathrm{kg\,m^{-3}}$

Remark

Unless otherwise stated, stellar quantities are quoted in solar-normalised form (for example $M/M_\odot$ and $L/L_\odot$), which makes scaling relations easier to compare across different stellar masses.

differential notation. We use $\dot{x} \equiv \frac{dx}{dt}$ for time derivatives of some variables $x(t)$.

CHAPTER 1

Basic Concepts in Astrophysics

Astrophysics is the branch of astronomy that applies the principles of physics and chemistry to understand the nature of celestial objects and phenomena. It includes but is not limited to the study of stars, galaxies, extrasolar planets, cosmic microwave background radiation, and the interstellar medium. In this notes, we will only focus on the field of stellar astrophysics, which is the study of stars, their properties, structure, types, and evolution.

1.1 Stars and Stellar Properties

Physics Intuition

In this notes, we define a star as a celestial object that is:

- (1) bounded by self-gravity under hydrostatic equilibrium, and
- (2) radiating energy generated by internal sources (e.g. nuclear fusion).

For most of the case, a star is spherical due to spherical symmetry of gravity, and mainly radiating energy by nuclear fusion in the core. However, we will also encounter some exceptions (e.g. energy generated by gravitational contraction in late stages of star evolution).

This definition shows that other celestial objects (e.g. planets, comets) are not stars as they don't fulfill all conditions. On the other hand, it allow some special conditions (e.g. brown dwarfs, white dwarfs) to be classified as stars, as they are still self-gravitating and radiating energy (though not by nuclear fusion). However, be careful some astrophysicists may not treat them as "real stars" as they don't have nuclear fusion, but we will still include them in our discussion as they are important for understanding the full picture of stellar evolution.

Remark

The stars are usually of spherical shape due to rotational symmetric force field of gravity, but can be oblate if they rotate rapidly.

1.1.1 Distance and Parallax

THEORY

- Parallax distance:

$$d(\text{pc}) = \frac{1}{\pi(\text{arcsec})} \tag{1.1}$$

- Radius from angular diameter θ:

$$R = \frac{1}{2}\theta d \tag{1.2}$$

Question Example 1.1

From the previous example, if the star has an angular diameter of 0.01 arcseconds, estimate its radius in units of $R_\odot$.

Solution.

$$\begin{aligned} R &= \frac{1}{2}\theta d = \frac{1}{2} \times 0.01'' \times 100\,\text{pc} \\ &= 0.5 \times 0.01 \times 100 \times 206265\,\text{AU} \\ &= 1031.325\,\text{AU} \\ &= 1031.325 \times 215 R_\odot \\ &\approx 221,000 R_\odot. \end{aligned}$$

1.1.2 Brightness, Magnitude, and Luminosity

THEORY

$$M_1 - M_2 = -2.5\log_{10}\left(\frac{L_1}{L_2}\right), \tag{1.3}$$

$$m_1 - m_2 = -2.5\log_{10}\left(\frac{F_1}{F_2}\right), \tag{1.4}$$

$$m - M = 5\log_{10}(d/\text{pc}) - 5, \tag{1.5}$$

$$F = \frac{L}{4\pi d^2}. \tag{1.6}$$

Question Example 1.2: Distance modulus quick use

A star has $m = 9.0$ and $M = 4.0$. Estimate its distance.

Solution.

$$\begin{aligned} m - M &= 5\log_{10}(d/\mathrm{pc}) - 5 = 5 \\ \Rightarrow \log_{10}(d/\mathrm{pc}) &= 2 \\ \Rightarrow d &= 10^2\,\mathrm{pc} = 100\,\mathrm{pc}. \end{aligned}$$

1.1.3 Mass

1.2 Thermal Radiation and Blackbody Spectrum

1.2.1 Planck function, Rayleigh-Jeans law, Wien's law

1.2.2 Intensity, Flux, and Luminosity

Physics Intuition

Observed starlight is measured in layers: direction-resolved intensity, surface flux, and finally total emitted power. Keeping these definitions separate avoids unit mistakes in later derivations.

Theory

- Specific intensity I_ν:

$$dE = I_\nu \cos\theta \, dA \, dt \, d\nu \, d\Omega \tag{1.7}$$

- Monochromatic flux density:

$$F_\nu = \int I_\nu \cos\theta \, d\Omega \tag{1.8}$$

- Bolometric flux:

$$F = \int_0^\infty F_\nu \, d\nu \tag{1.9}$$

- Luminosity:

$$L = \int F \, dA \tag{1.10}$$

Important Note

I_ν is invariant along a ray in vacuum (no absorption/emission), while F_ν depends on source geometry and distance. Do not interchange them in magnitude or distance calculations.

1.2.3 Radiative Transfer

Question Example 1.3: Calculating Brightness

Suppose a star is at 20 pc. How bright does it appear if its luminosity is $100L_\odot$?

Solution.

$$\begin{aligned} F &= \frac{L}{4\pi d^2} = \frac{100L_\odot}{4\pi(20\,\mathrm{pc})^2} \\ &= \frac{100 \times 3.828 \times 10^{26}\,\mathrm{W}}{4\pi(20 \times 3.086 \times 10^{16}\,\mathrm{m})^2} \\ &\approx 3.1 \times 10^{-9}\,\mathrm{W/m^2}. \end{aligned}$$

1.2.4 Surface Temperature

1.3 Surface Temperature and Spectral Classification

1.3.1 Blackbody Radiation and Wien's Law

THEORY

- Planck function:
$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/k_BT}-1} \tag{1.11}$$
- Stefan–Boltzmann law:
$$F = \sigma T^4, \quad \sigma = 5.67\times10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}} \tag{1.12}$$
- Luminosity:
$$L = 4\pi R^2\sigma T_{\mathrm{eff}}^4 \tag{1.13}$$
- Wien's law:
$$\lambda_{\max}T = 2.9\times10^{-3}\,\mathrm{m\,K} \tag{1.14}$$

Question Example 1.4

A star has a peak wavelength of 500 nm. Estimate its surface temperature.
Solution. From Wien's law,

$$\begin{aligned}\lambda_{\max}T &= 2.9\times10^{-3}\,\mathrm{m\ K}\\ T &= \frac{2.9\times10^{-3}\,\mathrm{m\ K}}{\lambda_{\max}}\\ &= \frac{2.9\times10^{-3}\,\mathrm{m\ K}}{500\times10^{-9}\,\mathrm{m}}\\ &= 5800\,\mathrm{K}.\end{aligned}$$

1.3.2 Spectral Classification

THEORY

Stars are classified into spectral types O, B, A, F, G, K, M based on their surface temperatures and spectral features. O-type stars are the hottest and most massive, while M-type stars are the coolest and least massive.

1.4 Star Formation

1.4.1 Hydrostatic Equilibrium

FOUNDATION

Assume spherical symmetry and radial dependence only. Neglect rotation, magnetic fields, and turbulence in the first model to isolate gravity-versus-pressure physics.

THEORY

$$dm(r,t) = 4\pi r^2 \rho \, dr, \tag{1.15}$$

$$\frac{\partial m}{\partial t} + v\frac{\partial m}{\partial r} = 0, \tag{1.16}$$

$$\frac{dP}{dr} = -\frac{Gm(r)\rho}{r^2} = -\rho g. \tag{1.17}$$

Question Example 1.5

Consider a sphere of mass M and radius R with a uniform density ρ.

(a) Write the pressure gradient $-\frac{dP}{dr}$ at distance r from the center at hydrostatic equilibrium.

(b) Calculate the pressure $P(r)$ at distance r from the center.

Solution.

(a) The mass enclosed within radius r is given by:

$$m(r) = \frac{4}{3}\pi r^3 \rho. \tag{1.18}$$

The hydrostatic equilibrium equation is:

$$-\frac{dP}{dr} = \frac{Gm(r)\rho}{r^2}. \tag{1.19}$$

Substituting $m(r)$, we get:

$$-\frac{dP}{dr} = \frac{G\left(\frac{4}{3}\pi r^3 \rho\right)\rho}{r^2} = \frac{4\pi G}{3} r\rho^2. \tag{1.20}$$

(b) To find $P(r)$, we integrate the pressure gradient from the surface ($r = R$, where $P(R) = 0$) to a point inside the sphere:

$$\begin{aligned} P(r) &= -\int_r^R -\frac{dP}{dr'}dr' = \int_r^R \frac{4\pi G}{3} r' \rho^2 dr' \\ &= \frac{4\pi G}{3}\rho^2 \int_r^R r' dr' = \frac{4\pi G}{3}\rho^2 \left[\frac{r'^2}{2}\right]_r^R \\ &= \frac{4\pi G}{3}\rho^2 \left(\frac{R^2 - r^2}{2}\right) = 2\pi G\rho^2(R^2 - r^2). \end{aligned}$$

1.5 Virial Theorem and Jeans Criterion

CHAPTER 2

Star Formation

2.1 Hydrostatic Equilibrium

FOUNDATION

Assume spherical symmetry and radial dependence only. Neglect rotation, magnetic fields, and turbulence in the first model to isolate gravity-versus-pressure physics.

2.2 Virial Theorem and Jeans Criterion

2.2.1 Energetics of Self-Gravitating Gas

THEORY

$$E_{\mathrm{GR}} = -\int_0^M \frac{Gm}{r}\,dm = -f_{\mathrm{GR}}\frac{GM^2}{R}, \tag{2.1}$$

$$\langle P\rangle = -\frac{1}{3}\frac{E_{\mathrm{GR}}}{V}, \tag{2.2}$$

$$P = \frac{1}{3}nm\langle v^2\rangle = \frac{2}{3}n\langle E_k\rangle. \tag{2.3}$$

For a star with a uniform density profile (i.e., $\rho = \text{constant}$), $f_{\mathrm{GR}} \approx 3/5$.

2.2.2 Jeans Instability

Theory

$$|E_{\text{GR}}| > 2E_{\text{th}} \quad \text{(collapse)}, \tag{2.4}$$

$$M_J = \left(\frac{5k_B T}{G\mu m_H}\right)^{3/2} \left(\frac{3}{4\pi\rho}\right)^{1/2}, \tag{2.5}$$

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G\rho}}, \tag{2.6}$$

$$M_J \propto T^{3/2}\rho^{-1/2}. \tag{2.7}$$

Important Note

Lower temperature and higher density both reduce M_J. This is why cooling strongly promotes fragmentation and cluster formation.

CHAPTER 3

Thermodynamic Properties of Stars

CHAPTER 4

Stellar Structure

Three main mechanisms of heat transfer in stars:

1. Conduction
2. Convection
3. Radiation (by photons and neutrinos from core)

CHAPTER 5

Nuclear Synthesis in Stars

Recall that there are three primary energy sources in stars: gravitational contraction, radioactive decay and nuclear fusion.

In this chapter, we will focus on the process of thermonuclear fusion, which is the viable long-term stellar energy source in most stars.

5.1 Nuclear Basics

5.1.1 Nuclear Size and Binding Energy

THEORY

$$R = R_0 A^{1/3}, \qquad R_0 \approx 1.2 \times 10^{-15}\,\mathrm{m}, \tag{5.1}$$

$$BE(Z,N) = \left[Z m_p + N m_n - M(Z,N)\right] c^2. \tag{5.2}$$

Binding energy per nucleon increases for light nuclei and decreases for very heavy nuclei due to Coulomb repulsion.

Question Example 5.1

Calculate the binding energy per nucleon for helium-4 (^{4}He) and ^{12}C.

Solution.

$$\begin{aligned} BE(^4\mathrm{He}) &= \left[2m_p + 2m_n - M(^4\mathrm{He})\right] c^2 \approx 28.3\,\mathrm{MeV}, \\ BE(^4\mathrm{He})/4 &\approx 7.1\,\mathrm{MeV/nucleon}, \\ BE(^{12}\mathrm{C}) &= \left[6m_p + 6m_n - M(^{12}\mathrm{C})\right] c^2 \approx 92.2\,\mathrm{MeV}, \\ BE(^{12}\mathrm{C})/12 &\approx 7.7\,\mathrm{MeV/nucleon}. \end{aligned}$$

5.1.2 Why Fusion Powers Stars

PHYSICS INTUITION

Stars gain energy when fusing light nuclei toward higher binding energy per nucleon. This trend reverses near iron, where further fusion is endothermic.

5.2 Hydrogen and Helium Burning

5.2.1 Proton-Proton (p-p) Chain

Foundation

Representative net reaction:

$$4\,{}^{1}\text{H} \to {}^{4}\text{He} + 2e^{+} + 2\nu_e + \text{energy}. \tag{5.3}$$

Total released energy is about 26.7 MeV, with part carried away by neutrinos.

5.3 CNO Cycle

5.4 Barrier Penetration and Reaction Rates

5.4.1 Coulomb Barrier and Tunneling

Derivation

Foundation

The **Gamov energy** is the minimum kinetic energy that two nuclei must have in order to overcome the Coulomb barrier and undergo fusion. It is given by the formula:

$$E_G = (\pi \alpha Z_1 Z_2)^2 2 m_r c^2 \tag{5.4}$$

where m_r is the reduced mass of the two nuclei, α is the fine-structure constant, and Z_1 and Z_2 are the atomic numbers of the two nuclei.

Question Example 5.2: The fusion of two protons

Calculate the Gamov energy E_G for the fusion of two protons.

Solution. For two protons, we have $Z_1 = Z_2 = 1$ and the reduced mass m_r is approximately equal to the mass of a proton, m_p. The fine-structure constant α is approximately $1/137$. Plugging these values into the formula for the Gamov energy, we get:

$$
\begin{aligned}
E_G &= (\pi\alpha Z_1 Z_2)^2 2m_r c^2 \\
&= (\pi \times \frac{1}{137} \times 1 \times 1)^2 \times 2m_p c^2 \\
&= \left(\frac{\pi}{137}\right)^2 \times 2m_p c^2 \\
&\approx 493\,\text{keV}.
\end{aligned}
$$

Thus, the Gamov energy for the fusion of two protons is approximately 493 keV.

Question Example 5.3: The fusion of carbon and oxygen

Calculate the Gamov energy E_G for the fusion of carbon-12 (^{12}C) and oxygen-16 (^{16}O).

Solution. For carbon-12, we have $Z_1 = 6$ and for oxygen-16, we have $Z_2 = 8$. The reduced mass m_r can be calculated using the formula:

$$
m_r = \frac{m_1 m_2}{m_1 + m_2}, \tag{5.5}
$$

where m_1 and m_2 are the masses of the carbon and oxygen nuclei, respectively. The mass of carbon-12 is approximately $12m_p$ and the mass of oxygen-16 is approximately $16m_p$. Thus, we have:

$$
\begin{aligned}
m_r &= \frac{(12m_p)(16m_p)}{12m_p + 16m_p} \\
&= \frac{192m_p^2}{28m_p} \\
&= \frac{192}{28} m_p \\
&\approx 6.86m_p.
\end{aligned}
$$

Now we can calculate the Gamov energy using the formula:

$$
\begin{aligned}
E_G &= (\pi\alpha Z_1 Z_2)^2 2m_r c^2 \\
&= (\pi \times \frac{1}{137} \times 6 \times 8)^2 \times 2 \times 6.86m_p c^2 \\
&= \left(\frac{48\pi}{137}\right)^2 \times 2 \times 6.86m_p c^2 \\
&\approx 8.7\,\text{MeV}.
\end{aligned}
$$

Thus, the Gamov energy for the fusion of carbon-12 and oxygen-16 is approximately 8.7 MeV.

FOUNDATION

Classically, nuclei with $E < V_C$ cannot fuse. Quantum tunneling gives non-zero penetration probability P for $E < V_C$. The probability of tunneling through the Coulomb barrier can be approximated by the formula:

$$P \propto \exp\left[-\sqrt{\frac{E_G}{E}}\right], \tag{5.6}$$

where E_G is the Gamow energy.

5.5 Thermonuclear Reaction Rates

THEORY

$$\langle \sigma v \rangle = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp\left[-\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}}\right] dE. \tag{5.7}$$

The integrand peaks at the Gamow energy window, balancing high-energy tail abundance and tunneling probability.

THEORY

$$R_{AB} \propto T^a \tag{5.8}$$

where $a = \sqrt[3]{\frac{E_G}{4kT}}$ is a positive exponent that depends on the specific reaction and the temperature range.

Question Example 5.4

Calculate the temperature dependence of the proton-proton (p-p) chain reaction rate.

Solution. The proton-proton chain reaction has a Gamow energy of approximately $E_G \approx 493\,\text{keV}$. The temperature dependence can be approximated by:

$$R_{pp} \propto T^{\sqrt[3]{\frac{E_G}{4kT}}}. \tag{5.9}$$

At typical stellar core temperatures of around $T \approx 1.5 \times 10^7\,\text{K}$, the exponent a can be calculated as:

$$a = \sqrt[3]{\frac{493\,\text{keV}}{4k(1.5 \times 10^7\,\text{K})}} \approx 4.5. \tag{5.10}$$

Thus, the reaction rate for the proton-proton chain scales approximately as $R_{pp} \propto T^{4.5}$ at typical stellar core temperatures.

CHAPTER 6

Simple Stellar Models

CHAPTER 7
Types of Stars

CHAPTER 8

Stellar Evolution

CHAPTER 9

Practice Questions & Solutions

9.1 Chapter 1: Basic Concepts in Astrophysics

9.1.1 Questions

1.1 The star Gliese 710 is presently 62 lightyears from Earth, but Hipparcos data suggest that in about a million years it will pass within one lightyear of Earth. Its appearent magnitude is 9.7.

   (a) What is the absolute magnitude of Gliese 710?

   (b) What will be its apparent visual magnitude be in a million years as viewed from Earth?

1.2 Estimate the hydrodynamical free fall timescale for the Sun, a red giant of $M = M_\odot$ and $R = 100R_\odot$, and a white dwarf of $M = M_\odot$ and $R = 0.01R_\odot$ and $R = 100R_\odot$, and a white dwarf of $M = M_\odot$ and $R = 0.02R_\odot$.

1.3 Suppose that a star of mass M and radius R has a density distribution $\rho(r) = \rho_c(1 - \frac{r}{R})$, where ρ_c is the density at the center of the star.

   (a) Find an expression for ρ_c in terms of M and R.

   (b) Calculate the mass $m(r)$ interior to radius r.

   (c) Calculate the total gravitational binding energy of the star.

   (d) Find an expression for the pressure $P(r)$ as a function of r, assuming that the star is in hydrostatic equilibrium and that the pressure at the surface $(r = R)$ is zero.

   (e) Assume that the material in the star is a **monatomic** ideal gas. Calculate the total internal energy of the star from $P(r)$, and show that the virial theorem is satisfied.

9.2 Chapter 2: Matter and Radiation

9.3 Chapter 3: Heat Transfers in Stars

9.4 Chapter 4: Thermonuclear Fusion in Stars

9.4.1 Questions

4.1 In the simplest model of α decay, the α particle is pre-formed and trapped within the nucleus by a potential similar to a Coulomb barrier. The mean decay rate λ is the frequency ν (with which the α particle hits the barrier) times the probability of penetration of the barrier (quantum tunneling probability).

(a) Write the approximate expression for the decay rate in terms of ν, E_G, and E (where E is the energy released by α decay).

(b) Find the Gamow energy of the α decay of ^{235}U.

(c) Find the Gamow energy of the α decay of ^{239}Pu.

(d) Assuming that the frequency ν is approximately the same for both nuclei, find the ratio

$$\frac{\lambda(^{239}\text{Pu})}{\lambda(^{235}\text{U})}$$

using $E(^{235}\text{U}) = 4.68\text{MeV}$ and $E(^{239}\text{Pu}) = 5.24\text{MeV}$. The lifetime of a radioactive species is inversely proportional to its decay rate. Given that the observed lifetime of ^{235}U is $\tau(^{235}\text{U}) = 7.1 \times 10^8 y$, estimate the lifetime of ^{239}Pu. Compare your result with the experimental value

$$\tau(^{239}\text{Pu}) = 2.41 \times 10^4 y$$

and comment on the accuracy of this simple model.

4.2 The **triple alpha process** is a set of nuclear reactions that occur in stars to produce carbon and other elements. Which of the following correctly describes the three steps of the triple alpha process, along with whether energy is absorbed or released during each step?

(a) $^4\text{He} +^4\text{He} \rightarrow^8 \text{Be}$

(b) $^8\text{Be} +^4\text{He} \rightarrow^{12} \text{C}^*$

(c) $^{12}\text{C}^* \rightarrow^{12} \text{C} + \gamma$

CHAPTER A

Declaration of AI Assistance

A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

CHAPTER B

Mathematical Tools & Derivations

B.1 Useful Identities

Trigonometric

$$\sin^2\theta + \cos^2\theta = 1, \quad \tan\theta = \sin\theta/\cos\theta, \quad \sin(A \pm B) = \sin A\cos B \pm \cos A \sin B.$$

Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Common Integrals

$$\int x^n\,dx = \tfrac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int e^{ax}\,dx = \tfrac{1}{a}e^{ax} + C,$$
$$\int \frac{dx}{x} = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C.$$

Gamma–Zeta Integrals

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\,dx = \Gamma(s)\zeta(s), \qquad \Re(s) > 1,$$
$$\int_0^\infty \frac{x^{s-1}}{e^x + 1}\,dx = (1 - 2^{1-s})\Gamma(s)\zeta(s), \qquad \Re(s) > 0.$$

$$\int_0^\infty \frac{x^2}{e^x - 1}\,dx = 2\zeta(3), \qquad \int_0^\infty \frac{x^2}{e^x + 1}\,dx = \tfrac{3}{2}\zeta(3),$$
$$\int_0^\infty \frac{x^3}{e^x - 1}\,dx = \frac{\pi^4}{15} = 6\zeta(4), \qquad \int_0^\infty \frac{x^3}{e^x + 1}\,dx = \frac{7\pi^4}{120} = \tfrac{7}{8}\,6\zeta(4).$$

Taylor Expansions

$$f(x) = f(a) + f'(a)(x-a) + \tfrac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$\begin{aligned}
\sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \tfrac{x^3}{3!} + \tfrac{x^5}{5!} - \cdots \\
\cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots \\
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \tfrac{x^2}{2!} + \cdots \\
\sqrt{1+x} &= \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \cdots
\end{aligned}$$

Common Approximations

Small-argument limits ($|x| \ll 1$):

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}, \qquad \tan x \approx x,$$
$$\sinh x \approx x, \qquad \cosh x \approx 1 + \frac{x^2}{2}, \qquad \tanh x \approx x.$$

Large positive argument ($x \gg 1$):

$$\sinh x \approx \frac{1}{2}e^x, \qquad \cosh x \approx \frac{1}{2}e^x, \qquad \tanh x \approx 1.$$

Large negative argument ($x \ll -1$):

$$\sinh x \approx -\frac{1}{2}e^{-x}, \qquad \cosh x \approx \frac{1}{2}e^{-x}, \qquad \tanh x \approx -1.$$

B.2 Relativistic Limits

The energy-momentum relation is $E^2 = p^2c^2 + m^2c^4$. So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

CHAPTER C

Constants & Conversion Factors

C.1 Physical Constants

Quantity	Symbol	Value (SI)
Speed of light	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
Hubble constant today	H_0	$67.8 \pm 0.77\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Cosmological constant	Λ	$1.11 \times 10^{-52}\,\mathrm{m^{-2}}$
Cosmological constant	Λ	$2.036 \times 10^{-35}\,\mathrm{s^{-2}}$
Planck constant	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s}$
Boltzmann constant	k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Fermi constant	G_F	$1.166 \times 10^{-5}\,\mathrm{GeV^{-2}}$
Fine-structure constant today	α	$1/137.036$
Planck mass	M_{Pl}	$2.176 \times 10^{-8}\,\mathrm{kg}$

C.2 Conversion Factors

Astronomical Units

Conversion	Value
1 astronomical unit (AU)	$1.496 \times 10^{11}\,\mathrm{m}$
1 light-year (ly)	$9.461 \times 10^{15}\,\mathrm{m}$
1 parsec (pc)	$3.086 \times 10^{16}\,\mathrm{m}$
1 parsec (pc)	$3.262\,\mathrm{ly}$
1 year (yr)	$3.156 \times 10^{7}\,\mathrm{s}$
1 Earth mass ($M_\oplus$)	$5.972 \times 10^{24}\,\mathrm{kg}$
1 Solar mass ($M_\odot$)	$1.989 \times 10^{30}\,\mathrm{kg}$
Milky Way' mass(M_{MW})	$1.5 \times 10^{12} M_\odot$
1 Earth radius ($R_\oplus$)	$6.371 \times 10^{6}\,\mathrm{m}$
1 solar radius ($R_\odot$)	$6.957 \times 10^{8}\,\mathrm{m}$
Milky Way' radius	$15\,\mathrm{kpc}$
1 solar luminosity ($L_\odot$)	$3.828 \times 10^{26}\,\mathrm{W}$
Milky Way' luminosity (L_{MW})	$3.5 \times 10^{10} L_\odot$

Energy Units

Conversion	Value
1 eV	1.602×10^{-19} J
1 electron mass (m_e)	$0.511\,\mathrm{MeV}/c^2$
1 electron mass (m_e)	9.109×10^{-31} kg
1 proton mass (m_p)	$938.3\,\mathrm{MeV}/c^2$
1 proton mass (m_p)	1.673×10^{-27} kg
1 neutron mass (m_n)	$939.6\,\mathrm{MeV}/c^2$
1 neutron mass (m_n)	1.675×10^{-27} kg

C.3 SI Unit Prefixes

Prefix	Name	Factor
10^{-12}	pico (p)	10^{-12}
10^{-9}	nano (n)	10^{-9}
10^{-6}	micro (μ)	10^{-6}
10^{-3}	milli (m)	10^{-3}
10^{3}	kilo (k)	10^{3}
10^{6}	mega (M)	10^{6}
10^{9}	giga (G)	10^{9}
10^{12}	tera (T)	10^{12}

C.4 Nuclei Binding Energies & Atomic Masses

Nucleus	Atomic Mass (u)	B(MeV)	B/A (MeV/u)
e^-	0.00054858	–	–
p	1.0072765	–	–
n	1.0086649	–	–
^{1}H	1.0078250	0.000000	0.000000
^{2}H	2.0141018	2.224515	1.112258
^{3}H	3.0160493	8.481773	2.827258
^{3}He	3.0160293	7.718036	2.572679
^{4}He	4.0026032	28.295803	7.073951
^{6}Li	6.0151223	31.994603	5.332434
^{7}Li	7.0160040	39.244654	5.606379
^{8}Be	8.0053051	56.499667	7.062458
^{9}Be	9.0121821	58.165096	6.462788
^{12}C	12.0000000	92.162051	7.680171
^{13}C	13.0033550	97.108223	7.469863
^{14}N	14.0030740	104.658962	7.475640
^{15}N	15.0001090	115.492214	7.699481
^{16}O	15.9949150	127.619412	7.976213
^{17}O	16.9991320	131.762631	7.750743
^{18}O	17.9991610	139.806972	7.767054
^{20}Ne	19.9924400	160.645559	8.032278
^{24}Mg	23.9850420	198.257479	8.260728
^{28}Si	27.9769270	236.537285	8.447760
^{32}S	31.9720710	271.781333	8.493167
^{56}Fe	55.9349420	492.255833	8.790283

Where **B** is the total binding energy of the nucleus, and **B/A** is the average binding energy per nucleon.

Note: $u = 1.661 \times 10^{-27}\,\text{kg} = 931.5\,\text{MeV}/c^2$.

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